Single-Turn Safety Evaluation Does Not Predict the Multi-Turn Robustness of Quantized Language Models

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Abstract

Quantization lets large language models run on edge hardware, where real use is multi-turn. Prior work measured quantization's effect on refusal at a single turn only and reported degradation. We give the first measurement across a multi-turn jailbreak, testing the claim on Llama-3.2-3B (FP16, NF4) and Llama-3.1-8B (FP16, AWQ, GPTQ) with both a refusal classifier and a Llama Guard 3 harm judge. We do not find degradation. On standalone AdvBench prompts, refusal is flat or higher under quantization for both models and all three methods, with no significant comparison. Under CoSafe multi-turn attacks, precision changes neither refusal (3B 23% vs 23%, McNemar $p = 0.62$; 8B 27 to 32%, Cochran Q $p = 0.12$) nor harmful-response rate; the one significant single-turn effect, 8B refusing more after quantization, vanishes under attack. A safety-prompt positive control confirms the harm metric can detect a real intervention, so the null is a true absence of effect. Along the way we find refusal-based attack success overstates harm by an order of magnitude (73% non-refusal versus 6% harmful). Single-turn safety evaluation does not predict how quantized models rank under conversational attack.

Introduction

Post-training quantization to 4 bits is now the standard way to fit an instruction-tuned model onto a phone or a laptop. The same deployment target that forces quantization also forces multi-turn use: a chat assistant holds a conversation. Safety evaluation has not followed. Recent work that measures how quantization changes refusal does so with single-turn prompts and reports a drop in refusal after quantization (Kadadekar 2026). Whether that drop holds, grows, or disappears once an attacker has several turns is open. The authors name multi-turn as future work.

We fill that gap. We ask one question: does 4-bit quantization change a model's multi-turn jailbreak robustness, and how does that compare to its single-turn effect on the same model and prompts? We answer it for two models and three quantization methods, and we read both behavior (refusal rate) and mechanism (projection onto the refusal direction of Arditi et al. (2024)).

Our finding is that the reported behavioral degradation does not reproduce, at either horizon. On standalone AdvBench prompts, quantization leaves single-turn refusal flat or higher for both models and all three methods, with no significant comparison. Under a CoSafe coreference attack (Yu et al. 2024), precision does not change refusal either; all precisions converge to a common non-refusal rate. The only significant behavioral precision effect anywhere is an *increase*: on the CoSafe cold turn, 8B quantization raises refusal, and even that vanishes under the attack. A Llama Guard 3 rescore confirms the null on a second, stricter metric: quantization never raises the rate of actually-harmful responses. The refusal direction still predicts refusal at every precision, so the models keep the representation.

The practical consequence is a warning about evaluation. A single-turn safety score does not preserve how FP16 and quantized checkpoints rank under conversational attack, and on these models it does not surface a behavioral safety cost from quantization at all.

Our contributions:

- A well-powered non-replication: across two Llama models, three 4-bit methods, two prompt sets (AdvBench, CoSafe), and two metrics (substring refusal and a Llama Guard harm judge), quantization does not reduce safety at a single turn.
- The first measurement of 4-bit quantization under a multi-turn attack: precision changes neither multi-turn refusal nor multi-turn harm, and the one significant single-turn effect does not survive the attack.
- An attribution decomposition of multi-turn attack success. Using the matched cold condition and a harm judge, we split the reported 73% success rate into 69 points of non-harmful output, 3 points of harm the model produces without any attack, and 2 points attributable to the attack, a 40x gap between what is reported and what the attack causes.
- A positive control showing the harm metric detects a real intervention (a safety prompt cuts multi-turn harm to zero), so the quantization null is a measured absence of effect rather than a lack of power.

Related Work

Quantization and safety. Sub-16-bit deployment is routine: LLM.int8() (Dettmers et al. 2022) handles the activation outliers that break naive schemes, and GPTQ (Frantar et al. 2023), AWQ (Lin et al. 2024), and NF4 (Dettmers et al. 2023) bring accurate post-training quantization to 4 bits, the setting we study. Safety behavior installed by preference training (Ouyang et al. 2022; Bai et al. 2022) and probed by red teaming (Ganguli et al. 2022; Perez et al. 2022) is brittle to post-hoc modification: Qi et al. (2024) show a few fine-tuning steps undo alignment even without malicious intent, and Egashira et al. (2024) construct models that behave benignly at full precision and turn harmful once quantized. Hong et al. (2024) audit trustworthiness across compression more broadly. Closest to us, Kadadekar (2026) asks whether retained output quality can substitute for direct safety testing after quantization. It cannot: quality stays flat while single-turn refusal falls 12 to 68 points in affected cells, and the direction of the effect splits across models rather than holding uniformly. Chhabra and Khalili (2025) separately report the refusal direction survives compression with cosine similarity above 0.99. All of this is single-turn, and Kadadekar (2026) lists multi-turn strategic attacks as out of scope and names them as future work. That is the gap we fill.

Jailbreak attacks. Single-turn attacks span gradient-based suffixes (Zou et al. 2023b) and stealthy prompt search (Liu et al. 2024; Chao et al. 2023), with Wei, Haghtalab, and Steinhardt (2023) attributing failures to competing objectives and mismatched generalization. Multi-turn attacks instead build harmful context gradually: Crescendo (Russinovich, Salem, and Eldan 2024) escalates over benign-looking turns, and CoSafe (Yu et al. 2024), which we use, hides the request behind coreference so the final turn is only legible in context. Li et al. (2026) show the projection onto the refusal direction decays across turns in full-precision models; we measure the analogous quantity to compare precisions rather than to characterize that decay.

Measuring jailbreak success. Scoring is itself contested. Many attack papers report success as the absence of refusal strings, which Souly et al. (2024) show overstates effectiveness relative to human judgment, proposing instead a rubric for how useful the response is. HarmBench (Mazeika et al. 2024) and JailbreakBench (Chao et al. 2024) standardize evaluation with trained classifiers, and model-based judges (Zheng et al. 2023; Inan et al. 2023) are now the common instrument. Our question is adjacent but distinct: prior critiques compare *judges* on single-turn attacks, asking which scorer best matches human labels. We hold the judge fixed and ask what a reported multi-turn success rate is composed of, using a matched no-conversation baseline to separate harm the attack caused from harm the model emits anyway, and from non-refusal carrying no harm at all. To our knowledge this attribution decomposition has not been applied to multi-turn attacks.

Method

Models and precisions. Both models are from the Llama 3 family (Dubey et al. 2024). Llama-3.2-3B-Instruct at FP16 and NF4 (bitsandbytes double-quantized). Llama-3.1-8B-Instruct at FP16, AWQ-INT4, and GPTQ-INT4, using the public `hugging-quants` checkpoints. AWQ and GPTQ layers are not fused, so forward hooks see the residual stream. FP16 extraction and ablation run on Apple silicon; quantized kernels run on CUDA.

Refusal direction. We extract the direction by difference-in-means (Arditi et al. 2024), the standard linear-probe construction in representation-level analyses of model behavior (Zou et al. 2023a): mean activation at the post-instruction token for harmful prompts minus the same for harmless prompts, per layer, normalized. Held-out separation accuracy is 100% for the 3B direction.

Causal layer by ablation. We do not pick the layer with the largest separation. We ablate each candidate layer's direction from the residual stream and keep the layer whose removal collapses refusal. For 3B this is layer 20 and for 8B layer 12: ablation drops refusal from 94% to 0% at that layer, while ablating a random direction leaves refusal at baseline. The separability-argmax layer is different in both models (28 for 3B, 31 for 8B) and does not mediate refusal, so we report only runs from the ablation-confirmed layer.

Attack measurement. For each CoSafe scenario we take the final (unsafe) user turn and measure it two ways. *Cold*: the final turn alone, no conversation. *In-context*: the full conversation, which is the attack. In both we record refusal, detected by a fixed phrase classifier on greedy generation, and the projection of the post-instruction activation onto the refusal direction. The cold and in-context conditions share the same final request, so their difference isolates the effect of the conversational context. Because CoSafe final turns use coreference, the cold condition is often under-specified on its own; this is the intended contrast, since the context is what makes the request actionable.

Standalone single-turn control. The CoSafe cold condition is a within-item control but not a standard harmful prompt. We add a second single-turn measurement on 100 standalone AdvBench (Zou et al. 2023b) instructions, presented as one turn, to test the quantization effect on overtly harmful prompts in the same setting as prior single-turn work. Same refusal classifier and projection readout.

Harm judge and positive control. Refusal only asks whether the model declined. To measure whether the response is actually harmful, we rescore every generated completion with Llama Guard 3 (Inan et al. 2023), reporting the harmful-response rate. As a positive control on this metric, we also generate every FP16 condition with a safety system prompt prepended, so an intervention known to change behavior gives the harm metric something to detect.

Statistics. Refusal is paired binary across precision, so we use McNemar's test per pair and Cochran's Q across the three 8B precisions. Projection is paired continuous, so we use the

Model	Prec.	Refusal		Refusal-dir. proj.	
		cold	in-ctx	cold	in-ctx
3B	FP16	25%	23%	$+1.46$	-0.86
3B	NF4	23%	23%	$+1.33$	-0.84
8B	FP16	23%	27%	$+0.88$	$+0.82$
8B	AWQ	31%	32%	$+0.80$	$+0.76$
8B	GPTQ	34%	31%	$+0.90$	$+0.89$

Table 1: Refusal and mean projection onto the refusal direction, cold (single turn) and in-context (multi-turn), $n = 100$ per cell.

Comparison	Cold	In-context
3B FP16 vs NF4	$p = 0.68$	$p = 0.62$
8B FP16 vs AWQ	$p = 0.043$	$p = 0.13$
8B FP16 vs GPTQ	$p = 0.010$	$p = 0.22$
8B Cochran Q	$p = 0.008$	$p = 0.12$

Table 2: Refusal significance (McNemar per pair, Cochran Q across the three 8B precisions). The single-turn precision effect on 8B is significant; the multi-turn effect is not.

paired t-test. We report the 100-scenario runs; $n = 100$ per condition.

Results

Multi-turn refusal does not depend on precision. Table 1 gives the rates, and Figure 1(a) plots the in-context refusal. For 3B the two precisions are identical at 23% (McNemar $\chi^2 = 0.25$, $p = 0.62$), and the paired projection difference is null ($t = 0.57$, $p = 0.57$). For 8B the three precisions span 27 to 32%, and Cochran's Q over the paired refusals is not significant ($Q = 4.2$, $p = 0.12$). The non-refusal rate sits near 70% for every precision, though most of that is not harmful output (harm results below).

The single-turn precision gap does not survive the attack. The cold column tells a different story for 8B. Quantization raises single-turn refusal: FP16 refuses 23%, AWQ 31%, GPTQ 34%. Both pairwise contrasts against FP16 are significant (McNemar $p = 0.043$ and $p = 0.010$; Table 2), and Cochran's Q is significant ($Q = 9.7$, $p = 0.008$). The same three precisions are not separable once the conversation is present: the 8 to 11 point spread at a single turn narrows to 5 points in context and loses significance (Table 1). 3B shows no precision effect at either horizon.

The direction stays coupled. If quantization broke the refusal representation, projection would stop predicting refusal. It does not. The AUC of projection predicting refusal ranges from 0.79 to 0.93 across every precision and condition, never approaching chance. Point-biserial correlations run from 0.46 to 0.68, all with $p < 10^{-6}$. The refusal direction drives behavior under AWQ, GPTQ, and NF4 as it does under FP16. Multi-turn convergence is therefore behavioral saturation across a shared threshold, not a loss of the mechanism.

Model	Refusal			McNemar
	FP16	quantized		vs FP16
3B	76%	NF4 82%		$p = 0.15$
8B	95%	AWQ 98%	GPTQ 95%	$p = 0.25/0.62$

Table 3: Single-turn refusal on 100 standalone AdvBench prompts. No precision comparison is significant, and quantization never lowers refusal.

Where the single-turn effect lives. The 8B single-turn caution gap concentrates in two CoSafe categories: privacy violation (FP16 40% cold, AWQ 52%, GPTQ 56%) and misinformation (FP16 4%, AWQ and GPTQ 16%). It is not a decoding artifact: garbled or empty output scores as non-refusal, so an artifact would lower the measured rate, not raise it, and the effect appears independently under two quantization methods.

No degradation on standalone harmful prompts. On AdvBench single-turn prompts (Table 3) quantization does not lower refusal either. 3B moves 76% to 82% under NF4 (McNemar $p = 0.15$). 8B moves 95% to 98% under AWQ ($p = 0.25$) and stays at 95% under GPTQ ($p = 0.62$). No comparison is significant, and every point estimate is flat or higher under quantization. On the two Llama models studied here we therefore do not observe the single-turn refusal degradation reported for INT4 quantization. We read this as consistent with Kadadekar (2026), who finds the effect splits across models rather than holding uniformly, and our result places both Llama checkpoints on the unaffected side. Refusal here is near ceiling for 8B, which limits behavioral power; the harm judge below addresses that from the other side. Quantization does produce a small, consistent shift in the projection onto the refusal direction (paired $t = -24.4$ for AWQ, -7.8 for GPTQ), but the shift is under 1% of the harmful-versus-harmless separation and does not predict which prompts flip, so we do not read it as a behavioral safety change.

The null holds under a harm judge, and refusal overstates harm. Refusal is a lenient signal. We rescore every completion with Llama Guard 3 (Inan et al. 2023), which labels whether the response itself is harmful (Table 4). Two things follow. First, quantization does not raise the harmful-response rate in any condition; every quantized run is at or below its FP16 baseline and no comparison is significant, so the null survives a second, stricter metric. Second, the harm rate is far below the non-refusal rate: the CoSafe attack takes 8B to a 73% non-refusal rate by the substring metric, but only 6% of those responses are judged harmful. The attack mostly produces non-refusing but non-harmful text. The gap is specific to the multi-turn setting: averaged over precisions, refusal-based attack success overstates harm by 4.9x at a single turn but 16.5x under the multi-turn attack, and of the responses the refusal metric flags as attacked, only 6% are actually harmful multi-turn versus 50% single-turn. A multi-turn jailbreak that reports success by non-refusal therefore measures whether the model kept talking, not whether it pro-

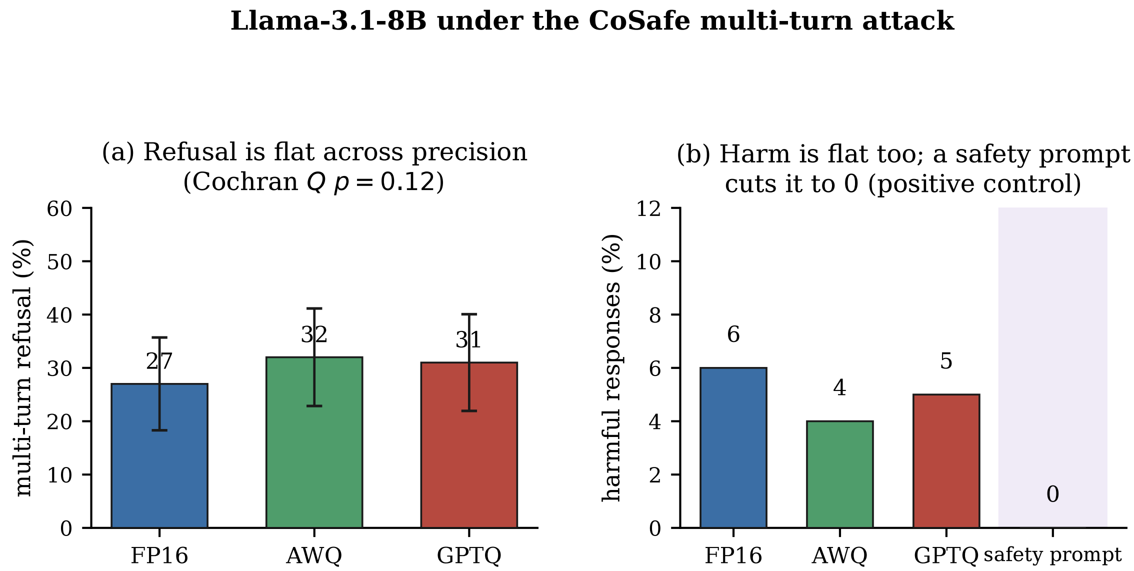

Figure 1: The two-metric null in one view. (a) Multi-turn refusal does not vary with precision (Wilson 95% intervals overlap). (b) Multi-turn harmful-response rate is low and does not rise under quantization, yet a safety system prompt drops it to 0%, showing the metric can detect a real intervention.

duced harm.

Decomposing reported attack success. The harm rate tells us the reported number is too high, but not what it is made of. The matched cold condition lets us separate the parts. A reported success is attributable to the attack only if the model refused the same request when asked cold, complied once the conversation was present, and produced harmful content. Compliance that also occurs cold is not evidence the attack worked, and non-refusal with no harmful content is not an attack success at all. Figure 2 gives the decomposition. Of the 73% reported success rate, 69 points produce no harmful content, a further 3 points reproduce harm the model also emits with no conversation, and 2 points are attributable to the attack. Reported multi-turn attack success therefore exceeds attack-attributable harm by roughly 40x, and the ordering across precisions is preserved under the stricter measure: quantization does not increase attack-attributable harm either.

The harm metric has power: a positive control. A null on a rare-event metric invites the objection that nothing could show up. To rule that out, we prepend a safety system prompt to every FP16 condition, an intervention that should change behavior. It does, and the harm metric detects it: under the multi-turn attack, the system prompt cuts 8B harmful responses from 6% to 0% (McNemar $p = 0.041$) and raises multi-turn refusal from 27% to 38% ($p = 0.006$; Figure 1b). So the harm metric is not saturated; it registers a real safety intervention on the same prompts where quantization registers nothing. The quantization null is a measured absence of effect, not an absence of power. The system prompt is the opposite of quantization here: it moves multi-turn safety, which is what makes it a clean control.

Setting	Model	FP16	quantized
single turn	3B	1%	NF4 1%
single turn	8B	4%	AWQ 2% / GPTQ 3%
multi turn	3B	4%	NF4 3%
multi turn	8B	6%	AWQ 4% / GPTQ 5%

Table 4: Harmful-response rate (Llama Guard 3), lower is safer. Quantization never raises it; all FP16-vs-quantized comparisons are non-significant (McNemar $p \geq 0.48$).

Discussion

Why the gap closes. CoSafe attacks work by hiding the harmful request behind coreference, so the model answers the earlier, benign-looking turns and then completes the final one in context. This attack is strong: it takes FP16 from 23 to 27% refusal for 8B and drives 3B to 23%. Once the attack is this effective, a few points of extra caution from quantization is not enough to change the outcome. The precision difference is real at a single turn and is swamped by the attack in context. A safety margin that exists in isolation does not imply a safety margin under pressure.

Implication for evaluation. Single-turn refusal is the standard axis for comparing a quantized checkpoint to its FP16 source. Our result shows that axis does not preserve the multi-turn ordering. For 8B, single-turn evaluation ranks the quantized checkpoints as *safer* than FP16, while under attack they are statistically tied. A single-turn number, whether it shows degradation or improvement, does not transfer to the conversational setting where the model is deployed.

Reported multi-turn attack success is 40x its attributable harm

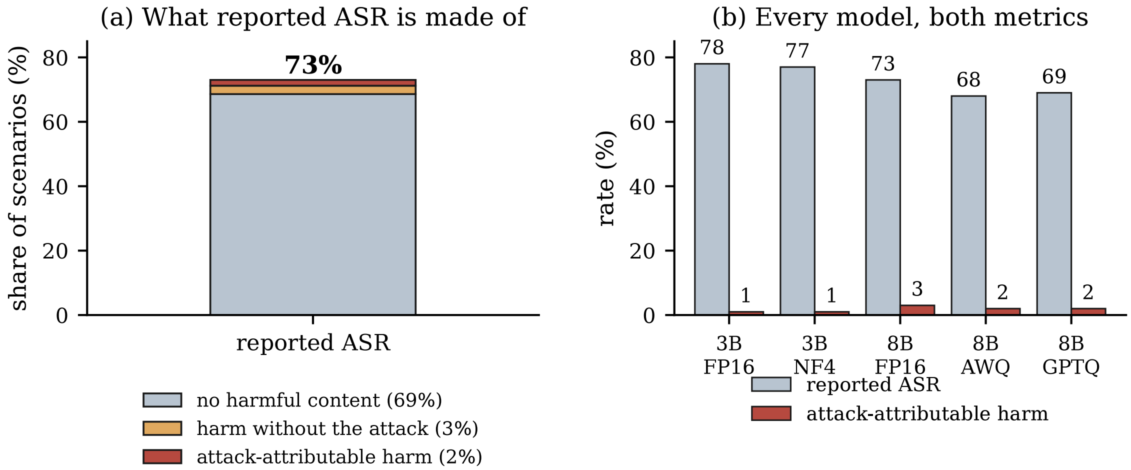

Figure 2: (a) The reported multi-turn attack success rate is almost entirely non-harmful output. (b) Reported success and attack-attributable harm for every model and precision.

Relation to prior reports. Kadadekar (2026) reports single-turn degradation whose direction splits across models rather than holding uniformly. Our two Llama checkpoints fall on the unaffected side: on AdvBench, the same standard single-turn setting, no precision comparison is significant and every point estimate is flat or higher under quantization, and on the CoSafe cold turn 8B quantization instead raises refusal. The claim here is therefore not that the earlier effect is wrong, but that it does not extend to the multi-turn setting for these models, and that a refusal-based multi-turn evaluation would have reported a 73% attack success rate that is almost entirely non-harmful. The Llama Guard rescore also closes the ceiling concern that would otherwise bound a refusal-only null: on a metric with room below (harm at 1 to 6%), and one the positive control shows can detect a real intervention, quantization still moves nothing.

Limitations

Two models and one attack dataset. Two model sizes (3B, 8B), one family (Llama), one multi-turn attack (CoSafe coreference), and three quantization methods (NF4, AWQ, GPTQ). We do not vary attack type or study training-time quantization. The refusal detector is a phrase classifier, which can miss soft refusals; the projection results, which do not use it, agree with the behavioral results. CoSafe safe categories only. The single-turn cold condition inherits CoSafe coreference, so it is a within-item control rather than a standalone harmful prompt; the AdvBench control noted above addresses that directly. Metric range. Substring refusal sits near ceiling (95% for 8B on AdvBench) and the Llama Guard harm rate near floor (1 to 6%), so each metric alone has limited room to expose a small effect. The positive control mitigates this for the harm metric, which still detects the system prompt, but the low harm rate partly reflects the CoSafe safe categories being mild relative to the severe-hazard taxonomy Llama Guard is tuned for, so it may under-flag them. The result is best read as: on these prompts, actual harmful output is rare at every precision, not as a guarantee on severe-harm categories.

Conclusion

Across two Llama models, three 4-bit methods, two prompt distributions, and two metrics, 4-bit quantization does not reduce safety at either a single turn or across a multi-turn attack. The only significant behavioral precision effect is an increase (8B on the CoSafe cold turn), and it does not survive the attack, where all precisions converge near 30% refusal and 4 to 6% harmful output. A safety system prompt, by contrast, does lower multi-turn harm on the same prompts, so the null is a true absence of effect, not a lack of measurement power. Single-turn safety evaluation does not predict how quantized models rank under conversational attack, and on these models it does not show a behavioral safety cost from quantization at all.

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